

JAVA Assignment

Assignment 6

Total Marks: 20

- Q1.** Write a Java program to design a **Smart Hospital Billing and Insurance Claim System** using abstract classes, abstract methods, interfaces, the final keyword (final variables, final methods and a final class) and runtime polymorphism. [20]

Scenario: A hospital treats three kinds of patients: General Ward patients, ICU patients and Out-Patients (OPD). Every patient pays a fixed registration fee and senior citizens get a discount on treatment. Admitted patients may claim part of their bill from insurance, but OPD patients cannot. The billing algorithm and the patient identity must be locked so that no subclass can tamper with them.

- (a) Interface InsuranceClaimable** [3]

- (i) Declare the constant double MAX_CLAIM_LIMIT = 100000.0; **without** writing public static final, and justify in a comment why it is still a constant.
- (ii) Declare the method boolean isEligibleForClaim();
- (iii) Declare the method double computeClaim(double grossBill); whose implementation must return 0 for an ineligible patient and must never exceed MAX_CLAIM_LIMIT.

- (b) Abstract class Patient** [5]

- (i) public static final constants REGISTRATION_FEE = 500.0, SENIOR_DISCOUNT_RATE = 0.10 and SENIOR_AGE = 60, and a counter private static int idCounter = 1000.
- (ii) A **blank final** field private final String patientId; which must not be initialised at declaration. The constructor must auto-generate it as P1001, P1002, P1003, and so on.
- (iii) protected fields name (String), age (int) and daysAdmitted (int), and a constructor Patient(String name, int age, int daysAdmitted).
- (iv) A **final** method public final String getPatientId().
- (v) A **final** method public final double generateBill() that computes: treatment = calculateTreatmentCost(); if age is 60 or more, reduce treatment by 10%; return REGISTRATION_FEE + treatment (the *Gross Bill*).
- (vi) Abstract methods public abstract double calculateTreatmentCost(); and public abstract String getWardType();

- (c) Concrete subclasses as per Table 1** [6]

Table 1: Subclass specifications

Class	Declaration	Additional fields	Treatment cost	Insurance rule
GeneralWardPatient	extends Patient implements InsuranceClaimable	final medicineCost Bed: Rs. 1,500/day	days x 1500 + medicineCost	Eligible if days >= 2; claim = 60% of Gross Bill
ICUPatient	final class , extends Patient, implements InsuranceClaimable	final ventilatorDays ICU: Rs. 8,000/day Ventilator: Rs. 5,000/day	days x 8000 + ventilatorDays x 5000	Eligible if days <= 30; claim = 90% of Gross Bill
OutPatient	extends Patient only (not insurable)	final consultationFee final diagnosticCost	consultationFee + diagnosticCost	Not applicable; daysAdmitted = 0

All per-day charges and claim percentages must be private static final constants. Every claim is capped at MAX_CLAIM_LIMIT. getWardType() must return "General", "ICU" and "OPD" respectively.

- (d) Driver class HospitalBillingSystem** [4]

- (i) Declare the Scanner reference as final. Read the number of patients n (re-prompt if $n \leq 0$) and create Patient[] patients = new Patient[n];
- (ii) For each patient, read the type (general / icu / outpatient; re-prompt if invalid), name (may contain spaces), age and the type-specific inputs. For an ICU patient, ventilator days must lie between 0 and days admitted; otherwise re-prompt. Create the appropriate subclass object at run time.
- (iii) Display a report with ID (via the final getter), Name, Ward, Gross Bill (via generateBill()), Claim, Net Payable (= Gross Bill - Claim) and Claim Status. Decide the status at run time using instanceof InsuranceClaimable and a cast: "Approved", "Not Eligible" or "Not Insurable".
- (iv) Finally display the total gross billing, total insurance claims, total amount collected and the patient with the highest net payable amount.

- (e) Restrictions imposed by final** [2]

Add the following fragments **inside comments** at the end of your program so that it still compiles. For each, write the exact compile-time error reported by javac and state in one line why Java forbids it.

- (i) A class FakeGeneral extends GeneralWardPatient that overrides generateBill() to return 0.
- (ii) A class SpecialICU extends ICUPatient.
- (iii) A method changeId(String newId) inside Patient that reassigns patientId.

Sample Input / Output (output layout may vary; computed values must match)

Enter number of patients: 3

--- Patient 1 ---

Type (general/icu/outpatient): general

Name: Rahul Das

Age: 45

Days admitted: 4

Medicine cost: 3200

--- Patient 2 ---

Type (general/icu/outpatient): icu

Name: Anjali Bora

Age: 65

Days admitted: 15

Ventilator days: 6

--- Patient 3 ---

Type (general/icu/outpatient): outpatient

Name: Kiran Kalita

Age: 30

Consultation fee: 800

Diagnostic cost: 2500

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===== HOSPITAL BILLING REPORT =====
ID      Name      Ward      Gross Bill      Claim  Net Payable  Claim Status
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P1001   Rahul Das   General    9700.00        5820.00    3880.00    Approved
P1002   Anjali Bora   ICU       135500.00     100000.00   35500.00    Approved
P1003   Kiran Kalita   OPD        3800.00         0.00     3800.00    Not Insurable
-----
Total Gross Billing      : Rs. 149000.00
Total Insurance Claims  : Rs. 105820.00
Total Collected (Net)  : Rs. 43180.00
Highest Net Payable     : Anjali Bora (P1002) - Rs. 35500.00
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